

Finite element analysis of punching shear behaviour of concrete slabs supported on rectangular columns

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ABSTRACT

Design provisions for punching shear of reinforced concrete flat slabs supported on rectangular columns vary greatly and are typically based on empirical results. Due to the large cost and time required to conduct experimental tests the empirical database for slabs supported on rectangular columns is small. Properly calibrated, nonlinear finite element models (FEM) provide an efficient way to expand experimental databases, conduct parametric studies and verify design provisions. In this paper, the results from a parametric study of the impact of column rectangularity (aspect ratio) on the punching shear behaviour of interior slab-column connections without shear reinforcement under static concentric vertical loading are provided. The study was conducted using a three-dimensional nonlinear FEM based on the concrete damaged plasticity model available in ABAQUS. The FEM, summarized herein, was calibrated using seven experimental specimens of slabs supported on square and rectangular columns from literature. The parameters studied were the column aspect ratio, and the ratio of the minimum column dimension to the effective slab depth. These parameters were selected based on a review of literature and design provisions. The impact of column rectangularity on the predicted load-deflection response, slab crack patterns and shear stress distribution in the slab along the column perimeter are analyzed. The results for the 77 slab-column connections simulated in the parametric study demonstrate that the impact of column rectangularity is not dependent on the column aspect ratio alone, and becomes more severe as the ratio of the minimum column dimension to the effective slab depth increases.

1. Introduction

Brittle punching shear failures of flat reinforced concrete slabs supported on columns occur due to the high shear stresses present in the area near the slab-column connection. Failure of the connection initiates from inclined cracks, which form a conical failure surface around the slab-column connection and extend from the tension surface to the compression zone of the slab. Brittle failure occurs when the compression zone can no longer sustain the applied loading. The study of punching shear is important due to the popularity of flat slab systems, and the fact that punching failure of one connection can lead to the progressive collapse of a structure.

To date, a large amount of research has been dedicated to understanding the mechanism of punching shear and the development of corresponding code design provisions. The existing code provisions are primarily based on empirical tests of isolated slab-column connections. Many of these testing programs are summarized in databases such as those compiled by the *fib* [1] and Joint ASCE-ACI Committee 445 [2]. However, due to the cost and time required to conduct experimental

studies, the current empirical databases cannot address all parameters which impact the punching shear behaviour of concrete slabs. The study of slabs supported on rectangular columns, which are commonly used in practice for architectural reasons and to reduce the slab clear span in the elongated column direction [3], has not been as extensively studied as slabs supported on square columns. For example, of the 519 tests currently included in the ASCE-ACI 445 database [2], only 27 are for slabs supported on rectangular columns. Paiva, Ferreira, Oliveira, Lima Neto and Teixeira [4] compiled one of the largest databases of experimental tests that included slabs supported on rectangular columns and included 68 such tests in the database that consists of 131 specimens. However, many of the tests in the database by Paiva et al. [4] are for one-way slabs subjected to concentrated loads.

Moe [5] was one of the first researchers to study the impact of column rectangularity on punching shear. He conducted a comprehensive test program which included 43 slabs but only one slab supported on a column with an aspect ratio of 3 was tested. Based on his tests, Moe proposed an equation to estimate the punching capacity of slab-column connections which was a function of the ratio of the

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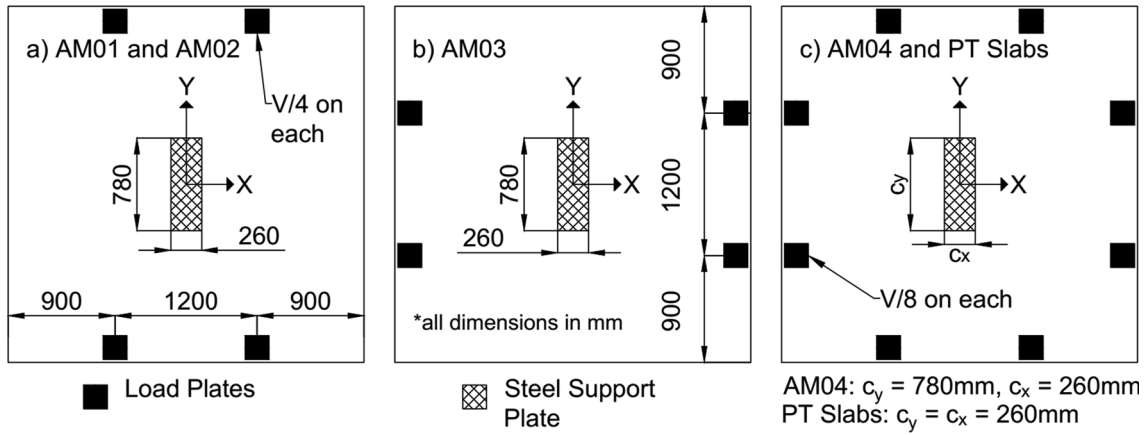


Fig. 1. Dimensions and load conditions in AM and PT series slabs [3,22], all dimensions in mm.

column size, c , to the effective slab depth, d , and considered the column perimeter as the critical perimeter. Moe also theorized that column rectangularity had a negative impact on punching capacity and proposed an equation to account for it but stated that further testing was required [5]. After Moe, Hawkins, Fallen and Hinojosa [6] tested 9 slabs supported on columns with aspect ratios between 1 and 4.33 under one-way and two-way loadings. They found that increasing the column aspect ratio led to a decrease in the nominal shear capacity along the assumed critical perimeter at $d/2$ from the column face. Other researchers (Sagaseta et al. [3], Teng et al. [7], and Oliveira et al. [8]) who have conducted experimental studies also confirmed the negative impact of column rectangularity on punching shear strength and the concentration of shear stresses along the short side and near the corner of the supported area (column).

Finite element analysis (FEA) is a cost- and time-effective way to supplement the existing experimental databases. First, using previous test results, finite element models (FEM) can be calibrated to accurately predict the punching capacity, failure mode, crack patterns and overall behaviour of the tested structural elements. Structural testing, due to the large scale involved, usually involves only a limited number of specimens with certain characteristics of specific interest. Therefore, the calibrated FEM can be helpful in conducting studies on various parameters affecting the behaviour of the structures under investigation. For example, Genikomsou and Polak [9] calibrated a three-dimensional nonlinear FEM in ABAQUS for the analysis of concrete slabs tested by Adetifa and Polak [10] and used it to study the impact of shear reinforcement [11,12], opening size and location [13] and compressive membrane forces [14] on punching shear behaviour. Shu et al. [15,16,17] have also developed finite element models based on both shell and continuum elements, which were used to predict the punching behaviour and internal force distribution in the slabs. Other researchers have shown that linear elastic FEA can be used to predict the shear stress distributions around rectangular columns [3,8,18] or to estimate the effective critical perimeter length [3,18].

This paper summarizes the calibration of a three-dimensional nonlinear FEM, based largely on the experimental program of Sagaseta et al. [3] which included four slabs supported on rectangular columns, and the results of a parametric study of the impact of column rectangularity on the behaviour of interior slab-column connections subjected to concentric punching. The calibrated model is based on the Concrete Damaged Plasticity (CDP) model available in ABAQUS, and the fictitious crack model by Hillerborg, Mod  er and Petersson [19], which considers the concrete fracture energy to reduce mesh sensitivity.

The parametric study concentrates on two parameters. The first parameter is the column rectangularity (aspect ratio), which is equal to the ratio of the maximum column dimension, c_{max} , to the minimum

column dimension, c_{min} . In this paper, this ratio is denoted as β ($\beta = c_{max}/c_{min}$) to be consistent with the ACI 318-19 provisions. The second parameter is the ratio of the minimum column dimension to effective slab depth, d , which is denoted κ in this paper ($\kappa = c_{min}/d$). These two parameters were considered based on a comparison of current and historical code provisions, discussed in detail in Milligan and Polak [20], and summarized herein. The trends observed in the predicted load-deflection response, crack patterns and shear stress distributions in the slab along the column perimeter considering 77 simulated isolated slab-column specimens are discussed in this paper. Finally, a discussion of the peak shear stress in the slab along the column perimeter, and a discussion of the increased influence of one-way shear behaviour for increased column sizes are provided.

2. Finite element simulations

The calibrated FEM by Genikomsou and Polak [9] served as the starting point for FEM used in this research. However, since different experimental specimens and test setups were considered, the FEM parameters required recalibration, as discussed by Eder et al. [21].

2.1. Experimental specimens from literature considered in calibration

Seven isolated slab-column test results were considered in the calibration of the FEM used in this research. Four of the seven slabs were from the AM series, which were tested by Sagaseta et al. [3] to study the impact of column rectangularity and load conditions on punching shear behaviour. The remaining three slabs were from the PT series, which were tested by Sagaseta et al. [22] to study the impact of non-axis symmetric reinforcing layouts on punching shear behaviour. As shown in Fig. 1, all seven slabs were 3000 mm square and 250 mm thick, but had different concrete strengths, reinforcing layouts and loading conditions. The slabs in the AM series were supported on rectangular steel plates with an aspect ratio of three (simulating the column support) [3], and the slabs in the PT series were supported on square steel plates [22]. The slabs were loaded through the square plates with different layouts as shown in Fig. 1.

Slab AM04 was the primary specimen used to calibrate the FEM since the focus of this research was the punching shear behaviour of slabs supported on rectangular columns subjected to concentric loading. Then the ability of the FEM to accurately account for different loading conditions and square columns was verified by modelling the remaining slabs in the AM series and select PT slabs, respectively. The material properties, reinforcing details, experimental results, and predictions from the calibrated FEM described herein, and in Genikomsou, Milligan and Polak [23] and Milligan and Polak [20] are provided in Table 1. A comparison of the load-rotation response predicted by the

Table 1
Summary of experimental program and FEA results.

Slab	f'_c MPa	d_{avg} mm	Tensile flexural reinforcement		Experimental results**				FEA Results	
			$\rho_x - \rho_y$ (%)	$f_{yx} - f_{yy}$ (MPa)	V_{exp} kN	ψ_x mRad	ψ_y mRad	V_{FEA} kN	ψ_x mRad	ψ_y mRad
AM01	44.0	214	0.75*	516–516	950	1.3	28.6	963.4	4.6	24.4
AM02	39.7	208	0.75*	516–516	919	5.2	26.3	926.0	5.0	20.6
AM03	42.2	203	0.75*	516–516	883	46.8	3.6	742.1	21.8	3.9
AM04	44.6	202	0.75*	516–516	1067	17.6	16.8	997.7	15.9	16.4
PT22	67.0	196	0.82–0.82	552–552	989	14.4	16.7	908.3	13.8	13.8
PT31	66.3	212	1.48–1.48	540–540	1433	9.5	11.6	1225.0	9.3	9.3
PT32	40.0	215	1.46–0.75	540–558	1157	10	12	961.4	8.9	12.2

*Only average flexural reinforcement ratio reported for AM series.

**Experimental rotations from AM series estimated from plots in Sagaseta et al. [3] average c_{min}/d ratio of AM series: 1.258, average c_{min}/d ratio of select PT slabs: 1.254.

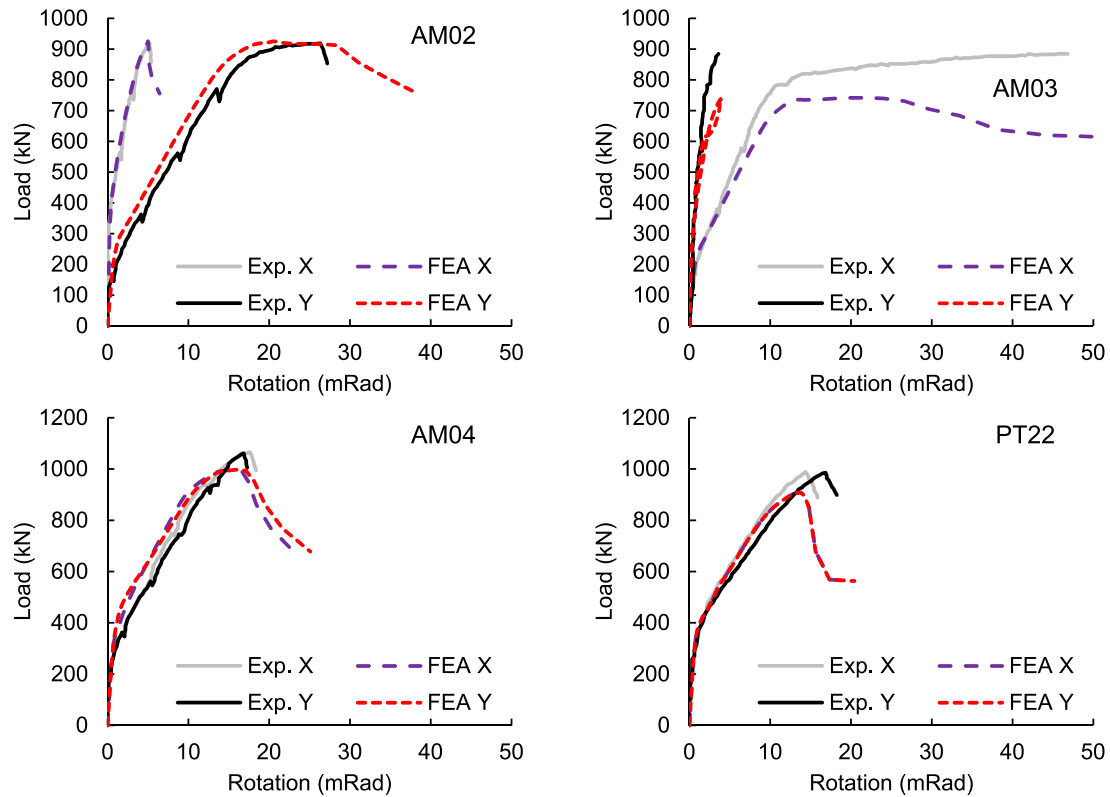


Fig. 2. Comparison of predicted and measured load-rotation response for AM02–AM04 and PT22.

FEM to the experimental results for slabs AM02, AM03, AM04 and PT22, which are representative of the overall trends can be found in Fig. 2. The experimental data provided in Fig. 2 for AM series and PT series are based on the figures provided in Sagaseta et al. [3] and Sagaseta et al. [22], respectively. A comparison of the predicted load-rotation responses for the remaining PT and AM series slabs can be found in Genikomsou et al. [23] and Milligan and Polak [20], respectively. As shown in Table 1 and Fig. 2, the calibrated FEM was able to accurately predict the behaviour and capacity of all slabs which experimentally failed in punching [3,22] (AM01, AM02, AM04, PT22, PT31 and PT32). For slab AM03, which failed in flexure [3], the predictions from the calibrated FEM begin to deviate from the experimental measurements at a load of approximately 600 kN. The likely cause for the discrepancy between the FEA results and the experimental measurements for slab AM03 is the use of a linear elastic perfectly

plastic stress-strain response for the flexural reinforcement. This simplified relationship was assumed in the FEM since the focus of this investigation was on the punching shear behaviour of slab-column connections and because the flexural capacity of slab AM04, which was the baseline slab for the parametric study described herein, was much higher than its punching capacity [3].

2.2. Summary of calibrated finite element model

ABAQUS/Explicit was used for the analyses [24]. To take advantage of symmetry, all slabs except for PT32, which was modelled as a half slab, were modelled as quarter slabs as shown in Fig. 3. Based on a boundary condition study, which is described in Genikomsou et al. [23], the steel load and support plates were included in the model, instead of applying the boundary conditions directly to the slab

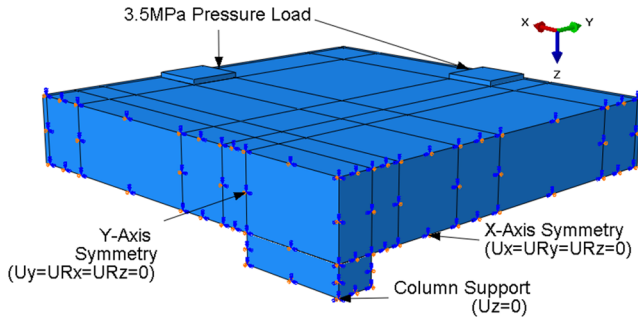


Fig. 3. Summary of boundary conditions in calibrated FEM.

geometry, to more accurately represent the experimental conditions where the slab was free to lift off the support [3,17]. Contact definitions between the slab and load and support plates were defined using the General Contact Algorithm in ABAQUS/Explicit. Between the base of the slab and the top of the support plate a “hard” normal contact, which allows for separation of the contact surfaces after initial contact, was used, and between the base of the load plates and top of the slab a “cohesive” contact, which does not allow separation of the contact surfaces, was used. In both cases, the defined contacts attempt to limit penetration of the contact surfaces. Pressure loads were applied to the top surface of the load plates to ensure uniform load application.

Three-dimensional, eight-node, reduced integration, hexahedral elements with default hourglassing controls (C3D8R) were used to mesh the concrete, support plate and load plates. The flexural reinforcement was modelled using three-dimensional, two-node, truss elements (T3D2) embedded in the concrete elements, assuming a perfect bond between the rebar and the concrete. A 20 mm global mesh size was used based on the results of a mesh sensitivity study [23], which resulted in 13 elements through the slab depth, which is larger than the minimum of 5 recommended for accurate results when using reduced integration elements [9].

The Concrete Damaged Plasticity (CDP) model available in ABAQUS was used to model the compressive and tensile behaviour of concrete. The Hognestad Parabola shown in Fig. 4 was used to model the uniaxial compressive behaviour of concrete for all slabs where f'_c was less than 60 MPa, which includes the 77 slabs in the parametric study. The post-peak response was terminated at a compressive stress of approximately 40% of f'_c to limit numerical issues caused by extending the post-peak curve to zero. For slabs PT22 and PT31, where f'_c exceeded 60 MPa, the uniaxial compressive stress-strain relationship by Thorendfeldt, Tomaszewicz and Jensen (1987), and valid for compressive strengths up to 125 MPa, was used [25].

Modelling of the uniaxial tensile stress-strain relationship for concrete was assumed to be linear elastic until the peak tensile stress, f'_t ,

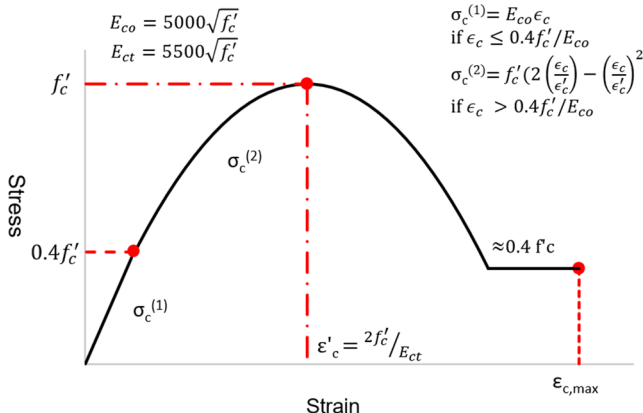


Fig. 4. Hognestad parabola assumed in calibrated FEM.

which was approximated using Eq. (1)

$$f'_t = 0.33\sqrt{f'_c} \quad (1)$$

Post-peak tension softening was accounted for using the bilinear tensile stress-crack width relationship proposed by Petersson [26], shown in Fig. 5. A tensile stress-crack width relationship was used to reduce the mesh sensitivity of the results, compared to directly inputting a tensile stress-strain relationship. Based on the calibration of the FEM discussed in [23] the fracture energy, G_f was calculated according to Model Code 1990 using Eq. (2)

$$G_f = G_{f0} (f'_{cm}/f'_{cm0})^{0.7} \quad (2)$$

where G_{f0} is the base fracture energy which is dependent on the maximum aggregate size (N/mm), f'_{cm} is the mean concrete compressive strength calculated as the characteristic concrete strength, f'_{ck} , plus 8 MPa, and f'_{cm0} equals 10 MPa [27]. The tensile stress-crack width relationship is then converted to a tensile stress-strain relationship in ABAQUS using the equations shown in Fig. 5 and a characteristic element length, l_c , which for C3D8R elements is equal to the cubic root of the element volume [9].

In addition to the uniaxial compressive and tensile stress-strain relationships, numerous other parameters are required in the CDP model in ABAQUS. One of the most influential parameters is the dilation angle. Based on a comparison of the load-rotation relationships and crack patterns predicted by the ABAQUS model with the experimental results provided in Genikomsou et al. [23], a dilation angle of 45 degrees was used for all models. The remaining CDP parameters are the eccentricity (ϵ), viscosity parameter (μ), ratio of initial equibiaxial compressive yield stress to initial uniaxial compressive yield stress (σ_{b0}/σ_{c0}), ratio of second stress invariant on the tensile meridian to the stress invariant on the compressive meridian (K_c), and Poisson's ratio (ν) of the concrete. The values for ϵ , σ_{b0}/σ_{c0} and K_c were set to their default values of 0.1, 1.16 and 0.667, respectively [24]. Since ABAQUS/Explicit was used for all analyses the viscosity parameter was not used. The Poisson's ratio was set to 0 to account for cracked concrete.

Finally, as previously mentioned, a linear elastic-perfectly plastic stress-strain relationship was assumed for the steel reinforcement since the focus of this work was on punching shear failures. The elastic modulus and Poisson's ratio of the flexural reinforcement were assumed as 200,000 MPa and 0.3, respectively. The yield strength of the flexural reinforcement was based on the properties reported by Sagaseta et al. [22] and Sagaseta et al. [3].

3. Parametric study

3.1. Selection of parameters

The material properties, slab size and experimental setup of slab AM04 tested by Sagaseta et al. [3] formed the basis of the slab-column connections analyzed in the parametric study. Therefore, the slabs in the parametric study were 3000 × 3000 mm in plan, 250 mm thick, had a concrete compressive strength of 44.6 MPa and were loaded at 8 discrete points through load plates (Fig. 6 shows the quarter model). As previously discussed, the parametric study was not focussed on studying the impact of the flexural reinforcement layout on punching shear behaviour, and as such, the slabs in the parametric study had the same flexural reinforcement layout and strength ($f_y = 526$ MPa [reinforcement on compression side] and 516 MPa [reinforcement on tension side]) as slab AM04 (see Table 1).

In order to conduct a rational parametric study on column rectangularity, proper selection of parameters had to be undertaken. To help with this task punching shear provisions specific to slabs supported on rectangular columns from Model Code 1978 [28], Model Code 2010 [29] and ACI 318-19 [30] were considered. Eurocode 2 (2004) was not considered as there are no provisions related to the impact of column

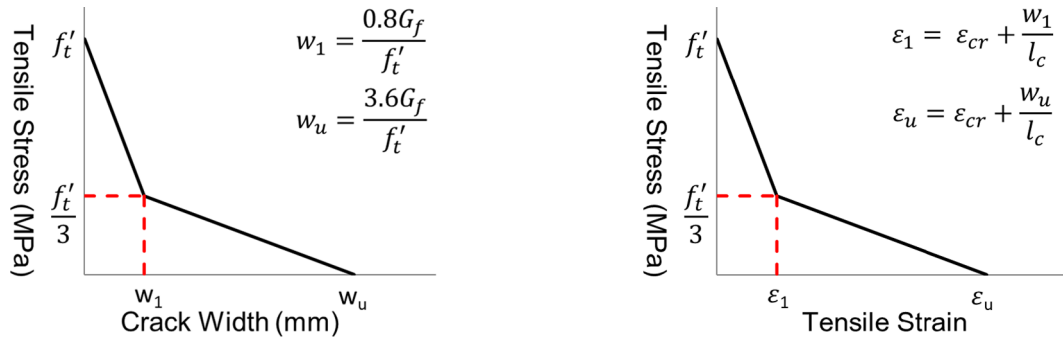


Fig. 5. Tensile stress-crack width relationship [26] and tensile stress-strain relationship [9] used in calibrated FEM.

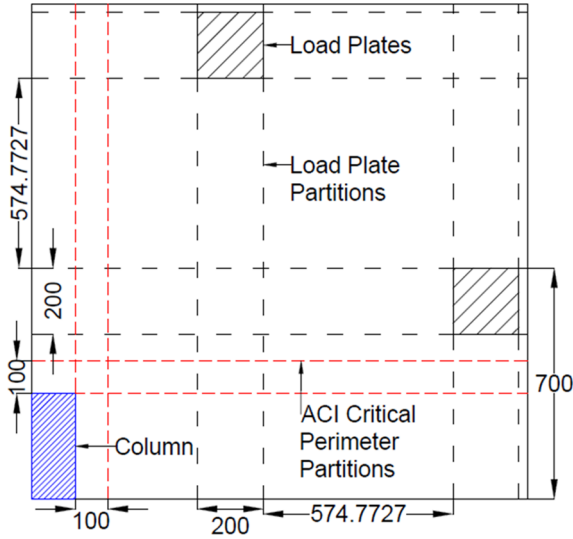


Fig. 6. Dimensions and layout of slab quarter model used in parametric study.

rectangularity on punching capacity for slabs subjected to concentric punching; in Eurocode 2 the use of a critical perimeter located at $2d$ from the column face results in an approximately uniform stress distribution for different column sizes [31]. To allow for comparison of the code provisions, a reduction factor for each code was calculated following the work of Sagaseta et al. [3] as discussed in the following paragraphs.

The ACI 318-19 provisions [30] assume a constant shear stress capacity along the entire critical perimeter length. For slabs with effective depths less than or equal to 250 mm (10 in), the nominal shear capacity, which is multiplied by the critical perimeter length and effective slab depth to estimate the punching capacity of a slab-column connection, can be expressed as some constant, denoted α in this work, multiplied by the square root of f'_c . For square columns where the ratio of the critical perimeter length, b_o , to the effective slab depth, d , is small (i.e. $< \approx 20.2$) α is equal to 0.33 (if f'_c is expressed in MPa). For rectangular columns with aspect ratios greater than 2, or columns where the b_o/d ratio is large, α is less than 0.33. Therefore, the reduction factor for ACI 318-19 is calculated as $\alpha/0.33$ to reflect the reduced nominal shear stress capacity around rectangular or large columns.

Model Code 1978 and Model Code 2010 account for column rectangularity by reducing the effective critical perimeter length, as shown in Fig. 7, instead of reducing the nominal shear capacity. Following the suggestion of Sagaseta et al. [3], the reduction factor for both Model Codes was calculated as the ratio of the effective critical perimeter length resisting punching, b_o , to the total critical perimeter length, b_1 .

The reduction factors for ACI 318-19, Model Code 1978 and Model Code 2010 for column rectangularities between 1 and 10 and for a slab with an effective depth of 200 mm are provided in Fig. 8 for κ equal to 0.6, 1.0 and 3.0. In Fig. 8 c_{max} and c_{min} denote the length of the longer and shorter sides of the column or supported area, respectively.

The impact of column rectangularity assumed in the two Model Codes is clearly dependent not only on the column aspect ratio, β ($\beta = c_{max}/c_{min}$), but also on column size with respect to the effective slab depth κ ($\kappa = c_{min}/d$). The ACI 318-19 predictions are nearly identical for all three κ values shown. The reduction factors for all three design codes were found to be similar for column rectangularities between 1 and 5 when κ was equal to 1, which is common in practice (Fig. 8b) [3]. For very small columns ($\kappa = 0.6$ in Fig. 8a) Model Code 2010 was found to predict column rectangularity to have a negligible impact on punching capacities for small to moderate rectangularities (i.e. less than 6). For large columns ($\kappa = 3$ in Fig. 8c) the Model Codes predict a much larger reduction in punching strength than the ACI code.

Based on the above analysis column rectangularity was studied considering two parameters: (1) the column aspect ratio, β (c_{max}/c_{min}), and (2) column size with respect to slab depth, represented by κ (c_{min}/d). As summarized by ASCE-ACI Committee 426 [32] many researchers have proposed punching shear design equations which are functions of the ratio of column size to the effective slab depth. The influence of this parameter has also been shown more recently by Al-Yousif and Regan [33], Oliveria et al. [8] and Einpaul et al. [34]. A summary of the connections simulated in the study is provided in Table 2. The column dimensions and rectangularity values for all 77 specimens are provided in Appendix A.

In Table 2, the difference between the desired and investigated κ values is due to the use of the AM04 geometry and 20 mm elements, which provided some constraints on the considered column sizes.

It should be noted that the κ values were also selected to be within the range of those in the ASCE-ACI 445 database [2]. Additionally, due to the use of a simplified stress-strain response for the flexural reinforcement the influence of the flexural reinforcement ratio was not included in the parametric study. None of the 77 slab-column connections analyzed were predicted to fail in flexure.

3.2. Parametric study - results

The observed trends in the predicted load-displacement relationships, crack patterns, and shear stress distributions in the slab were consistent for all κ values. Representative figures are provided in this paper and figures for other κ values can be found in Milligan [35]. In all discussions, the Y direction is parallel to the long side of the column and the X direction is parallel to the short side of the column as shown in Fig. 1 [3].

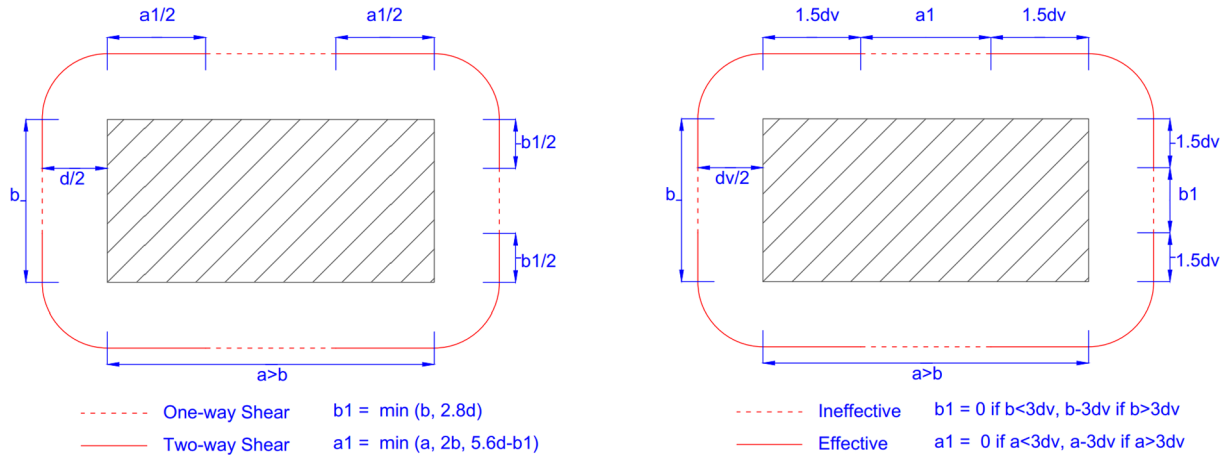


Fig. 7. Effective critical perimeters assumed in Model Code 1978 [28] (Left) and Model Code 2010 [29] (Right).

3.2.1. Load-displacement behaviour

The predicted load-displacement relationship for the specimens where κ equals 0.990 (close to 1) are shown in Fig. 9. For all κ values the FEM typically predicted an increased shear capacity as the column rectangularity, and the column size, increased, which is an expected result. No clear trends were observed in the predicted deflections at failure, but the slab-column connections were typically found to be stiffer in the direction along the long side of the column, which matches experimental results by numerous researchers [36,37,38]. The stiffness of the predicted response after the elastic region was found to increase as column rectangularity was increased and the stiffness in the Y direction was found to increase at a higher rate. As seen in Fig. 9, the predicted response in the X direction appears to become more ductile as the column rectangularity is increased. However, this pseudo ductility is likely due to the increased dominance of one-way shear behaviour as the longer column dimension increased.

The majority of the investigated specimens failed in punching. However, one-way shear failures, which initiated and occurred parallel to the long side of the column, were typically predicted when the maximum column dimension was approximately greater than or equal to 40% of the slab length of the isolated specimens, $L_{isolated}$, as shown in Table 3. Analysis of continuous slab systems by Simmonds [39] and Sagaseta et al. [3] have shown that one-way shear behaviour dominates when the ratio of the maximum column dimension, c_{max} , to the slab span between column centerlines, $L_{continuous}$, is between 0.35 and 0.4.

Table 2

Summary of slabs analyzed in parametric study.

κ (c_{min}/d ratio)		# of studies	Min. β (c_{max}/c_{min})	Max. β (c_{max}/c_{min})	Maximum column size (mm)
Desired	Actual				
0.5	0.594	19	1.0	10.000	120 × 1200
0.75	0.792	15	1.0	8.000	160 × 1280
1	0.990	12	1.0	6.600	200 × 1320
1.25	1.287	9	1.0	5.077	260 × 1320
1.5	1.485	7	1.0	4.000	300 × 1200
1.75	1.782	6	1.0	3.556	360 × 1280
2	1.980	5	1.0	3.000	400 × 1200
3	2.970	3	1.0	2.000	600 × 1200

$f'_c = 44.6$ MPa, $d_{avg} = 202$ mm, $\rho_t = 0.75\%$, 16 mm diameter bars ($f_y = 516$ MPa at 125 mm o/c both directions (tension side), 12 mm diameter bars ($f_y = 526$ MPa at 125 mm o/c both directions (compression side), slab thickness = 250 mm, slab properties and reinforcing match slab AM04 [3].

The one-way shear failures predicted in the current work support the conclusion that one-way shear behaviour dominates when the maximum column dimension is equal to approximately 40% of the slab span length. This behaviour also demonstrates the difference between slab-column connections in isolated specimens and continuous slabs when the $c_{max}/L_{continuous}$ ratio is less than 0.2 where one-way shear failures are predicted.

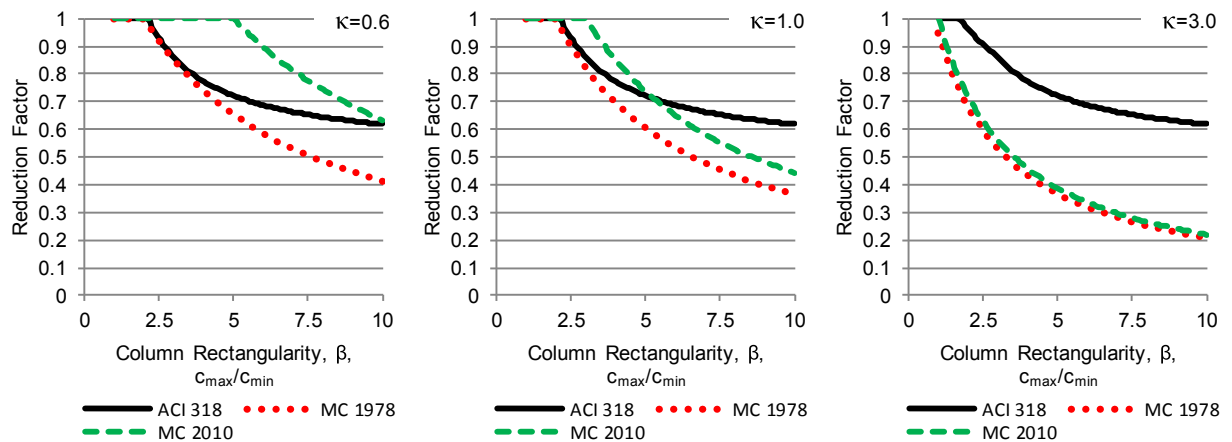


Fig. 8. Comparison of reduction factors from ACI 318-19, Model Code 1978 and Model Code 2010.

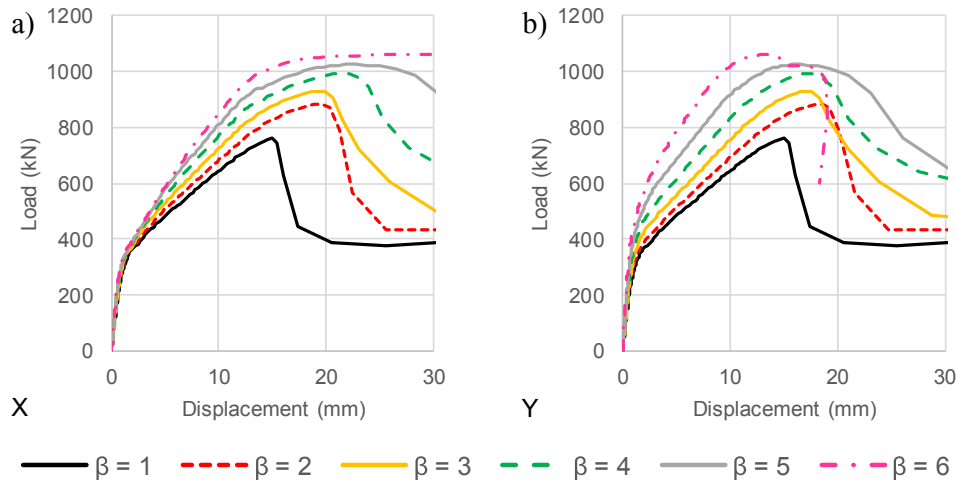


Fig. 9. Predicted load-displacement response, $\kappa = 0.990$.

Table 3

Minimum column rectangularities β for which one-way shear failures are predicted for each $\kappa = c_{min}/d$.

$\kappa = c_{min}/d$	β	c_{min} (mm)	c_{max} (mm)	$c_{max}/L_{isolated}$ ^(a)	$c_{max}/L_{continuous}$ ^(b)
0.594 ^(c)	10	120	1200	0.40	0.178
0.792	7.5	160	1200	0.40	0.178
0.990	6	200	1200	0.40	0.178
1.287	4.462	260	1160	0.39	0.172
1.485	4	300	1200	0.40	0.178
1.782	3	360	1080	0.36	0.160
1.980	3	400	1200	0.40	0.178
2.970 ^(d)	–	–	–	–	–

^a Length of isolated specimen = 3000 mm.

^b Length of continuous slab estimated as 6750 mm based on tests by Guandalini, Burdet and Muttoni [40], which are geometrically similar to AM04.

^c Simultaneous two-way and one-way shear failure predicted for $\beta = 10$ model.

^d One-way shear failure not found to govern for $\kappa = 2.970$ results. $\beta = 2$ model experienced secondary one-way shear failure mode.

3.2.2. Cracking

The trends in the predicted crack patterns on the tension surface and slab sides were consistent for all values of κ . Fig. 10 provides examples of the crack patterns on the tension surface and slab sides, visualized in ABAQUS through the contours of maximum principal plastic strain, for a slab with a square column and a slab with a rectangular column. The cracks are approximately the same in both perpendicular directions near the column for the slabs supported on square columns. As the column rectangularity is increased the crack patterns become more non-uniform: the diagonal cracking on the tension surface concentrates near the shorter side of the column and the cracks perpendicular to the column sides appear (Fig. 10i and iv).

The shape of the punching cone changes with increasing column rectangularity. For more rectangular columns the conical crack extending from the tension surface of the slab to the compression surface decreases in thickness as seen in the decreased maximum principal plastic strain magnitude on both sides of the slab (Fig. 10). Also, as the column rectangularity is increased the conical shape on the slab side perpendicular to the long side of the column transitions into a reverse Z shape, as shown in Fig. 10iii and vi. The lack of punching cone on the slab side perpendicular to the long side of the column supports the conclusion that one-way shear behaviour becomes more dominant as the column aspect ratio is increased. The increased dominance of one-

way shear is also supported by the formation of a one-way shear crack on the tension surface parallel to the long side of the column which grows in length before extending across the entire slab width as β was increased for each value of κ .

3.2.3. Shear stress distribution along column perimeter

The shear stress distribution in the slab around the column perimeter at the peak load (100%) for selected analysed specimens with a κ value of 1.287 are shown in Fig. 11. These shear stress distributions are representative of those observed for all investigated κ values and were calculated based on the deformed element areas and shear stresses at the element centroids. The deformed element areas were calculated considering the nodal coordinates of the four nodes on the outer vertical face. The shear stress at the element centroid was then multiplied by the deformed area to find the shear force carried by each element. Finally, for each vertical band of elements along the column perimeter the total shear force was divided by the total area of elements to determine the shear stress distribution along the column perimeter. A detailed discussion of the methodology used to calculate the shear stress distributions from the ABAQUS results is provided in [35].

For slabs supported on square columns, the predicted shear stress distributions were approximately uniform (though there was a shear stress concentration near the column corner), whereas for slabs supported on rectangular columns a concentration of shear stress was observed at the corners and along the short side of the column. Along the long side of the column, the predicted shear stress magnitudes were found to rapidly decrease away from the column corner. The shear stress magnitudes at a set distance from the corner of the columns were also found to be similar for all investigated specimens within each κ value, especially along the long side of the columns.

The shear stress distributions along the column perimeter were also analyzed at 30%, 90% and 100% of the peak load to study the shear distribution in the linear elastic portion of the response and close to failure, as shown in Figs. 12 and 13 for κ values of 1.287 and 1.782, respectively. Similar plots for all remaining κ values can be found in [35]. The shape of the predicted shear stress distribution changed with the applied load level. In the linear elastic portion of the response (30% of peak load) the concentration of shear stress near the corner and along the short side of the column is visible. The magnitude of the shear stresses along the short side of the column were found to be almost uniform for κ values less than 1.485. For larger columns with κ values greater than 1.485, the shear stresses were found to become increasingly concentrated at the corner of the column.

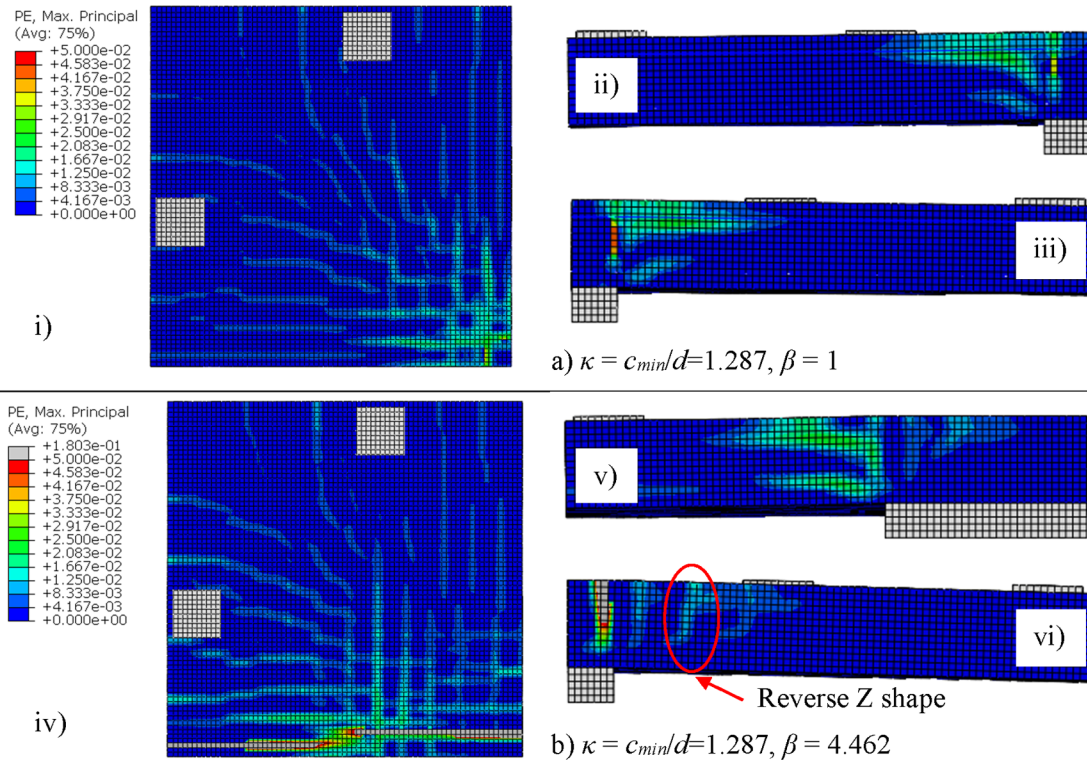


Fig. 10. Examples of predicted crack patterns for square ($\beta = 1$) and rectangular ($\beta = 4.462$) columns, for constant $\kappa = 1.287$, tension surface (i and iv), slab side parallel to long side of column (ii and v), slab side parallel to short side of column (iii and vi).

A shear stress redistribution from the short side to the long side of the column was observed between 90% and 100% of the peak load. After 90% of the peak load, the shear stresses along the short side of the column were typically found to decrease, whereas the shear stress magnitudes along the long side, especially near the column centerline increased. Clearly, failure is initiated by damage on the short side of rectangular columns with stresses being then redistributed to the long side of the column.

The peak shear stresses in the slab along the column perimeter, which were typically found to occur at the corner of the column, calculated from the ABAQUS results at 30%, 90% and 100% of the peak load level are provided in Fig. 14.

At 30% of the peak load the peak shear stresses were between 1.3 and 2 MPa, and typically increased as the column aspect ratio increased. The high peak shear stresses compared to the stresses along the

majority of the column perimeter at 30% of the peak load, as shown in Figs. 12 and 13 demonstrate that regardless of load level a large portion of the shear stresses in the slab are carried by the concrete near the column corner. The peak shear stresses for the slabs supported on square or rectangular columns are similar at 30% of the peak load. At 90% and 100% of the peak load, the peak shear stresses for the slabs supported on square columns are typically higher than those for the slabs supported on rectangular columns.

At 90% and 100% of the peak load level the peak shear stresses typically fall between 4 and 7 MPa as seen in Fig. 14, with a majority of the peak shear stress magnitudes falling between 5 and 6 MPa. This narrow range of peak shear stress values suggests that shear failures were initiated when the peak shear stress exceeded these values, which are within the range of 9–16% of f_c or 59–106% of $\sqrt{f_c}$, based on a f_c value of 44.6 MPa. Furthermore, at 100% of the peak load, the

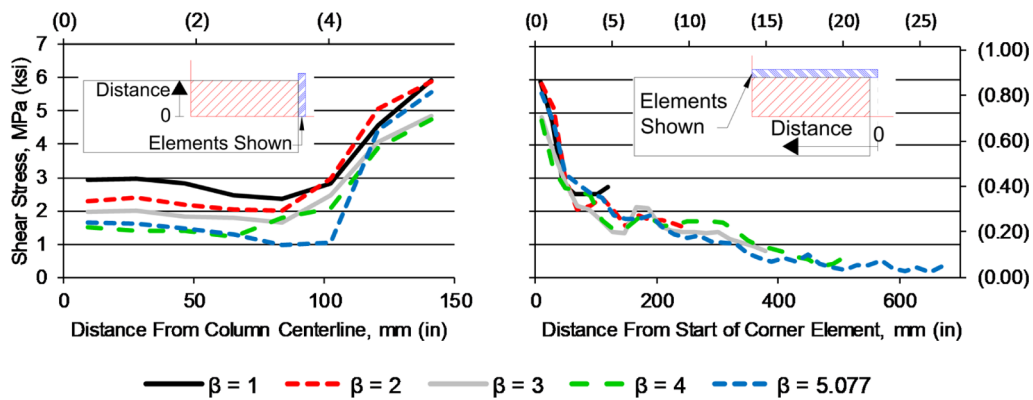


Fig. 11. Typical shear stress distributions along column perimeter, $\kappa = c_{min}/d = 1.287$, Left: short side of column (Centerline at Left), Right: long side of column (Centerline at Right), $1/4$ of column shown.

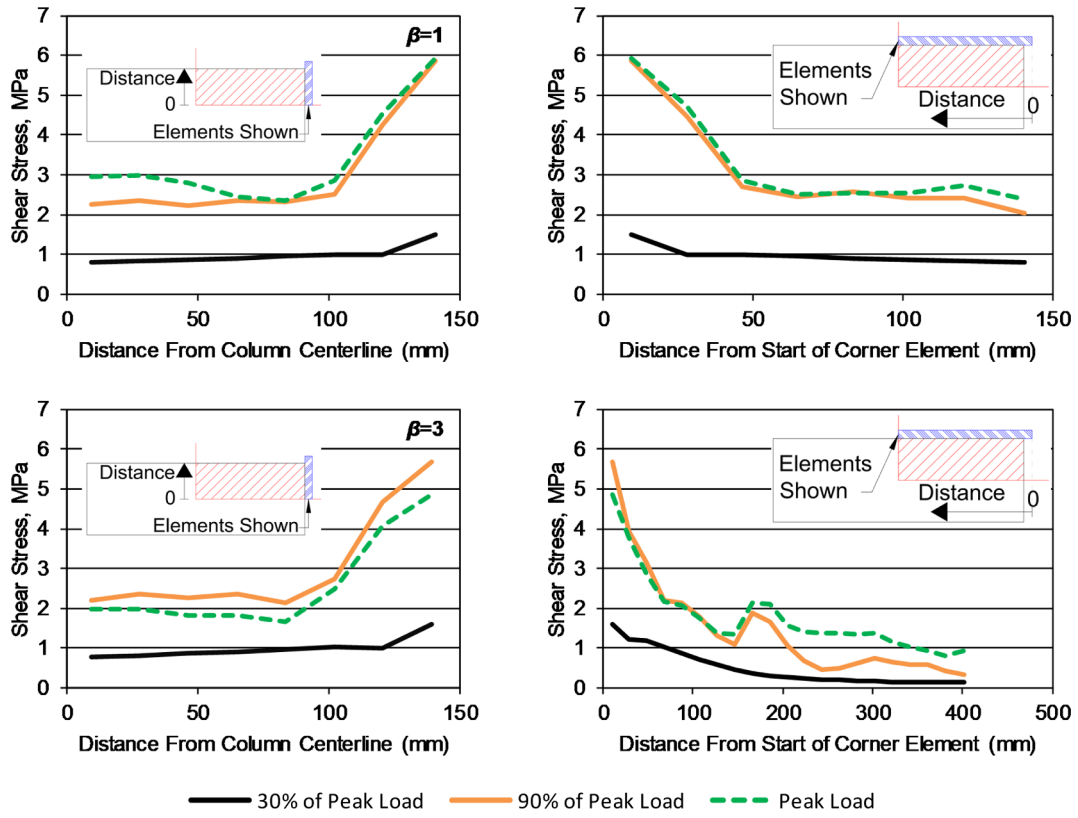


Fig. 12. Examples of change in shear stress distribution along column perimeter with load level, $\kappa = c_{min}/d = 1.287$, Left: short side of column (Centerline at Left), Right: long side of column (Centerline at Right), $\frac{1}{4}$ of column shown.

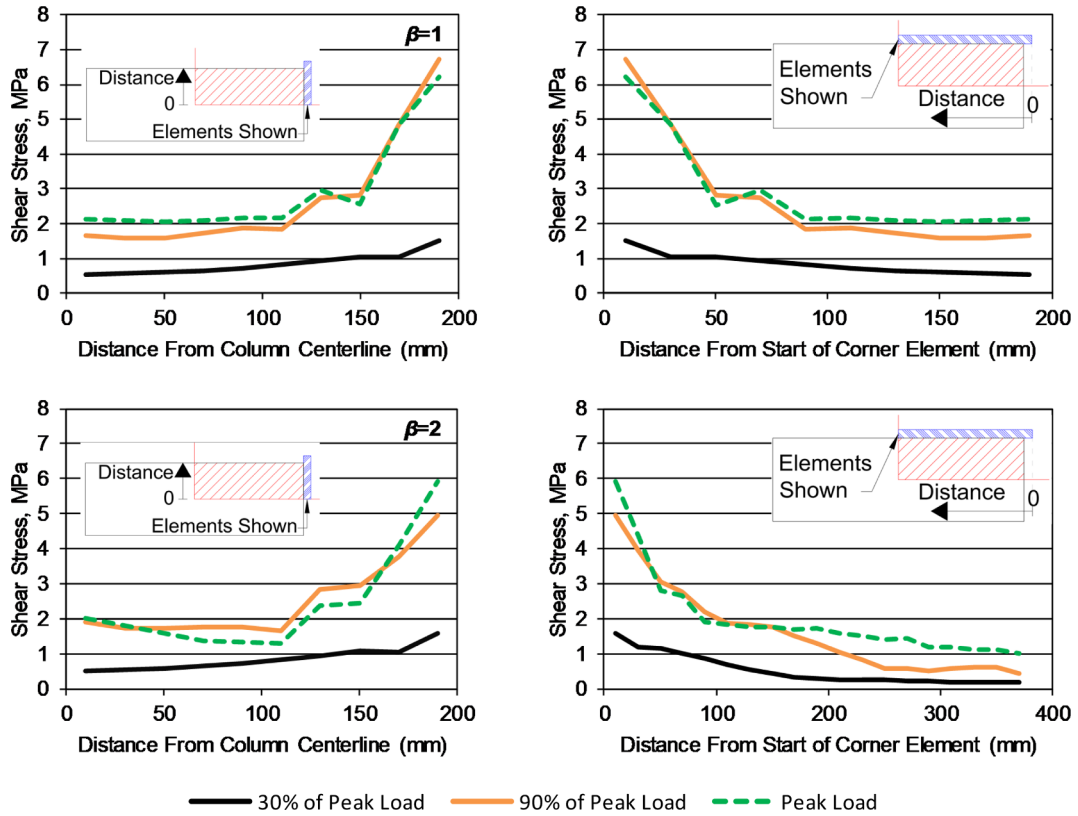


Fig. 13. Examples of change in shear stress distribution along column perimeter with load level, $\kappa = c_{min}/d = 1.782$, Left: short side of column (Centerline at Left), Right: long side of column (Centerline at Right), $\frac{1}{4}$ of column shown.

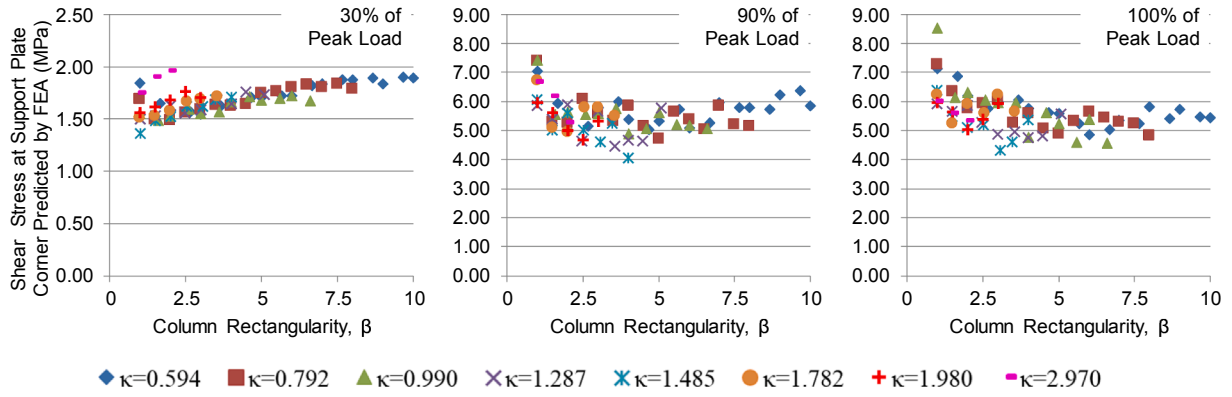


Fig. 14. Peak shear stress predicted in slab along column perimeter at 30%, 90% and 100% of peak load.

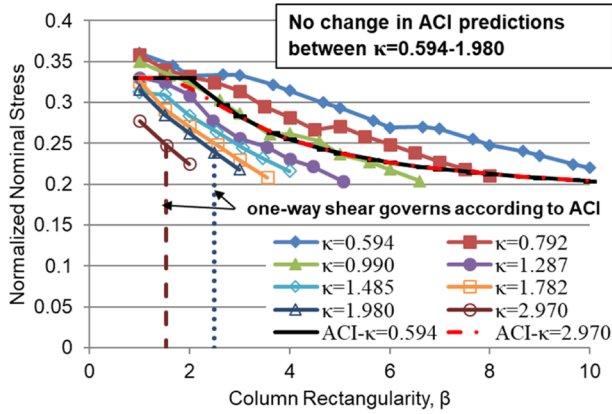


Fig. 15. Comparison of normalized nominal shear stress capacity along critical perimeter predicted by FEA and ACI 318-19.

predicted peak shear stresses appear to decrease to an approximately constant value as the column aspect ratio is increased for each κ value.

3.2.4. Comparison to ACI 318-19 provisions for punching shear

To compare the FEA predictions to the ACI 318-19 provisions a normalized nominal shear stress capacity along the ACI critical perimeter, v_{norm} , based on the FEA results was calculated using Eq. (3)

$$v_{norm} = V_{FEA} / (b_{o,ACI} d \sqrt{f_c}) \quad (3)$$

where V_{FEA} is the shear capacity predicted by the calibrated FEM (N), $b_{o,ACI}$ is the length of the critical perimeter according to ACI 318-19 (mm), d is the effective slab depth (mm) and f_c is the concrete compressive strength (MPa). For ACI 318-19 v_{norm} is equivalent to the reduction factor, α , presented in Section 3.1, and is calculated using Eq. (4) for slabs with effective depths less than or equal to 250 mm cast from normal density concrete

$$v_{norm,ACI} = \alpha = \min \left\{ \begin{array}{l} 0.33 \\ 0.17(1 + 2/\beta) \\ 0.083 \times (2 + \alpha_s \times d/b_o) \end{array} \right\} \quad (4)$$

where α_s is a constant dependent on the column location (equal to 40 for interior columns) [30]. It should be noted that for all square columns v_{norm} predicted by ACI 318-19 is 0.33 unless the ratio of the critical perimeter to effective slab depth is very large ($b_o/d > \approx 20.2$).

A comparison of the normalized nominal shear stress capacity along the critical perimeter calculated from the FEA results and according to ACI 318-19 for the 77 slab-column specimens considered in the

parametric study is provided in Fig. 15.

From Fig. 15 it is clear that the impact of column rectangularity predicted by the FEM is strongly dependent on the value of κ , and becomes more severe as κ increases as seen in the decreasing v_{norm} values for the same β . The ACI 318-19 provisions, on the other hand, are largely independent of κ and the value of v_{norm} predicted for a constant β is identical for all κ values less than or equal to 1.980. Only for the largest κ value investigated, 2.970, is a slight decrease in v_{norm} predicted for column aspect ratios between 1.5 and 3 according to the ACI 318-19 punching provisions. However, it should be noted that the ACI 318-19 one-way shear provisions govern when β is greater than to 2.5 and 1.533 for κ equals 1.980 and 2.970, respectively.

The relationship between v_{norm} and β predicted by the FEA for each κ value is almost approximately linear, whereas the relationship assumed in ACI 318-19 is curved. If a trendline is fit to the FEA predictions for each κ value the slope of the trendline is found to become increasingly negative as κ is increased, further demonstrating that the FEA predicts a more severe impact of column rectangularity for increasing κ . Finally, it is observed that the FEA results predict an impact of column rectangularity for column aspect ratios between 1 and 2, whereas ACI 318-19 assumes a constant value of v_{norm} .

4. Conclusions and discussions

In this paper, a three-dimensional nonlinear finite element model, implemented in ABAQUS using the Concrete Damaged Plasticity model and calibrated based on experimental results, was used to analyze the impact of column rectangularity on the punching shear behaviour of interior slab-column connections without shear reinforcement subjected to concentric vertical loading. Based on a comparison of code provisions the influence of two parameters, namely the column aspect ratio, β ($\beta = c_{max}/c_{min}$), and the ratio of the minimum column dimension, c_{min} , to the effective slab depth d , denoted as κ in this work ($\kappa = c_{min}/d$), on the shear behaviour of slab-column connections was studied. In total, 77 isolated slab-column connections with 8 κ values were analyzed. The main conclusions of the presented work are:

- (1) ABAQUS analysis results show an impact of column rectangularity on punching shear behaviour, which correlates well with that observed experimentally by other researchers. During the calibration ABAQUS' ability to accurately capture the influence of different loading and slab reinforcement layouts on the behaviour of isolated slab-column connections with square or rectangular columns was verified, particularly for connections which failed in shear.
- (2) The results of the parametric study demonstrate that the impact of column rectangularity on the punching shear behaviour of slab-

column connections is not dependent on the column aspect ratio only. It is also strongly influenced by the ratio of column size to slab depth, which is expressed as the ratio c_{min}/d (κ). As κ was increased the impact of column rectangularity on the nominal shear capacity along the critical perimeter at $d/2$ from the column face was found to become more severe. For each κ value, the relationship between column rectangularity and nominal shear stress capacity was found to be approximately linear, and the slope of the trendline fit to the predictions for each κ value was found to become more negative as κ increased.

- (3) An impact of column rectangularity on punching shear was predicted even for small rectangularities ($\beta \leq 2$), which does not match the assumption in ACI 318-19.
- (4) The stiffness of the slab-column connection in both orthogonal directions was found to increase with increasing column rectangularity. The stiffness in the direction along the long side of the column (the stiffer direction) was found to increase at a higher rate than the stiffness in the direction along the short side of the column.
- (5) One-way shear behaviour became dominant at lower column rectangularities as κ increased. However, the maximum column dimension where one-way shear failures were predicted were similar for all $\kappa = c_{min}/d$ ratios.
- (6) The shear stresses in the slab along the column perimeter were found to concentrate at the corner and along the short side of the supported area. The concentration of shear stresses at the corner of the supported area was found to increase as κ increased, whereas the concentration along the short side decreased.
- (7) The peak shear stresses at 90% and 100% of the peak load for the slabs supported on rectangular columns typically fell between 4 and 7 MPa with the majority of the peak shear stresses falling between 5 and 6 MPa. This narrow range of peak stresses suggests that punching failures occur at a consistent peak shear stress. In this work, the peak stresses were between 9 and 16% of f'_c or 59–106% of $\sqrt{f'_c}$, based on a f'_c value of 44.6 MPa.
- (8) The shape of the shear stress distribution along the column perimeter is dependent on the applied load level.

5. List of notation

b_o – length of critical perimeter according to ACI 318-19 or effective critical perimeter assumed in Model Code 1978 or 2010 (mm)	b_l – total critical perimeter length according to Model Code 1978 or 2010 (mm)	c_{max} – longer dimension of column/supported area (mm)
c_{min} – shorter dimension of column/supported area (mm)	d – effective slab depth (mm)	f'_c – concrete compressive strength (MPa)

Appendix A. Summary of specimens considered in parametric study

See Table A1.

f'_t – concrete tensile strength (MPa)	f_{ck} – characteristic compressive strength (MPa)	f_{cm} – mean compressive strength (MPa), equal to $f_{ck} + 8$ MPa
f_y – yield strength of flexural reinforcement (MPa)	G_f – concrete fracture energy (N/mm)	G_{f0} – base concrete fracture energy according to Model Code 1990 (N/mm) [27]
K_c – ratio of second stress invariant on tensile meridian to the second stress invariant on compressive meridian [24]	$L_{isolated}$ – slab length in isolated specimen (mm)	$L_{continuous}$ – centerline distance between columns in continuous slab system (mm)
V_{exp} – experimentally measured capacity (kN)	V_{FEA} – specimen capacity predicted by FEA (kN)	β – column aspect ratio (c_{max}/c_{min})
ϵ – eccentricity parameter in Concrete Damaged Plasticity Model [24]	ϵ_{cr} – concrete cracking strain, f_t/E_c	$\kappa = c_{min}/d$ ratio, relative column size
μ – viscosity parameter in Concrete Damaged Plasticity Model [24]	ν – Poisson's ratio	ν_{norm} – normalized nominal shear stress along ACI 318-19 critical perimeter
ρ_i – flexural reinforcing ratio in direction i	σ_{b0}/σ_{c0} – ratio of initial equibiaxial compressive yield stress to initial uniaxial compressive yield stress [24]	ψ_i – slab rotation in direction i (mRad)

CRedit authorship contribution statement

Graeme J. Milligan: Conceptualization, Methodology, Software, Validation, Investigation, Writing - original draft, Visualization. **Maria Anna Polak:** Conceptualization, Methodology, Validation, Resources, Writing - review & editing, Supervision, Project administration, Funding acquisition. **Cory Zurell:** Conceptualization, Methodology, Validation, Writing - review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table A1
Column dimensions and aspect ratios in parametric study.

$c_{min}/d = 0.594$			$c_{min}/d = 0.990$			$c_{min}/d = 1.782$		
c_{min}	c_{max}	β	c_{min}	c_{max}	β	c_{min}	c_{max}	β
120	120	1	200	200	1	360	360	1.000
120	200	1.667	200	320	1.6	360	540	1.500
120	240	2	200	400	2	360	720	2.000
120	320	2.667	200	520	2.6	360	920	2.556
120	360	3	200	600	3	360	1080	3.000
120	440	3.667	200	720	3.6	360	1280	3.556
120	480	4	200	800	4			
120	560	4.667	200	920	4.6			
120	600	5	200	1000	5	$c_{min}/d = 1.980$		
120	680	5.667	200	1120	5.6	c_{min}	c_{max}	β
120	720	6	200	1200	6	400	400	1.000
120	800	6.667	200	1320	6.6	400	600	1.500
120	840	7				400	800	2.000
120	920	7.667				400	1000	2.500
120	960	8	$c_{min}/d = 1.287$			400	1200	3.000
120	1040	8.667	c_{min}	c_{max}	β			
120	1080	9	260	260	1.0			
120	1160	9.667	260	390	1.5	$c_{min}/d = 2.970$		
120	1200	10	260	520	2.0	c_{min}	c_{max}	β
			260	640	2.462	600	600	1.000
			260	780	3.0	600	920	1.533
			260	920	3.538	600	1200	2.000
			260	1040	4.0			
$c_{min}/d = 0.792$								
c_{min}	c_{max}	β						
160	160	1	260	1160	4.462			
160	240	1.5	260	1320	5.077			
160	320	2						
160	400	2.5						
160	480	3	$c_{min}/d = 1.485$					
160	560	3.5	c_{min}	c_{max}	β			
160	640	4	300	300	1.0			
160	720	4.5	300	450	1.5			
160	800	5	300	600	2.0			
160	880	5.5	300	760	2.533			
160	960	6	300	920	3.067			
160	1040	6.5	300	1040	3.467			
160	1120	7	300	1200	4.00			
160	1200	7.5						
160	1280	8						

All dimensions given in mm.

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